

UNIT-1

Communication System transfers the data or information-bearing signal from source to destination via communication channel. Communication system is of basically two type. an analog or digital type.

- ① In analog type, the information-bearing signal is continuously varying in both amplitude & time, and it is used to modify some characteristics of a sinusoidal wave, such as amplitude, phase or frequency.
- ② In digital type, the information bearing signal is processed & it is represented by a sequence of discrete messages.

Digital communication system is being popular & preferred over analog communication system due to following reasons.

- * Lot of development made in the field of computer, which can be used as a digital source. There by Telex and facsimile are being digitized.
- * Digital communication system is preferred due to its flexibility and compatibility.
- * The reliability is being improved in digital communication system over analog comm. system.
- * The digital communication system is capable of handling wideband channels provided by GEO Satellites, optical fibre cables and co-axial cables.
- * Developments made in the field of IC Technology, a very complicated digital circuit can be designed and fabricated in a single IC chip, which reduces the cost of the system.

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THIS NOTES AVAILABLE



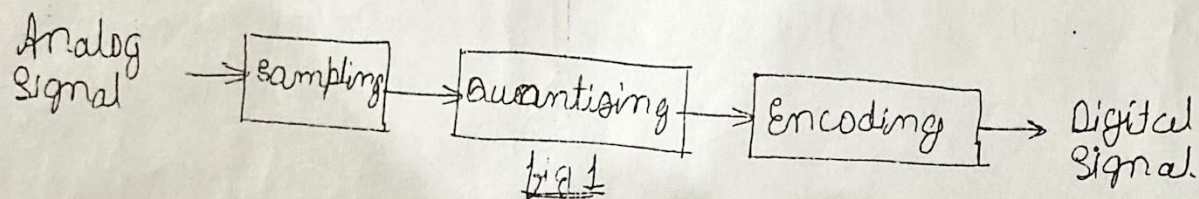
SRI SAI MERON
Opp: M.L. Nagar, 103
145342, HUDA, Gurgaon

Source and Signals

The source of information generates a message, but it is not electrical in nature and a transducer is used to convert it into an electrical waveform called message signal, or "baseband signal" means band of frequencies representing the message signal generated at the source.

The message signal can be of an analog or digital type. In the analog signal both amplitude & time vary continuously over their respective intervals e.g.: speech signal, television signal. In digital signal both amplitude and time take on discrete values. e.g.: computer data, telegraph signal.

An Analog signal can be converted into digital form by three basic operations: Sampling, Quantizing & Encoding as in fig 1.



In Sampling process, only sample values of the analog signal at uniformly spaced discrete instants of time are retained. In quantizing operation, each sample value is approximated by the nearest level in a finite set of discrete levels. In the encoding operation, the selected level is represented by a code word that consists of a prescribed number of code elements. Fig 2 shows a segment of analog waveform.

In code word shown in fig 3 consists of four binary digits, with the last bit assigned the role of a sign bit that signifies whether the sample value is positive or negative, the remaining three bits represents the binary representation of quantized levels.

As a result of sampling and quantizing operations, errors are introduced into the digital signal. These errors are non-reversible, so that it is not possible to produce an exact replica of the original analog signal from its digital representation, by proper selection of the sampling rate and code word length.

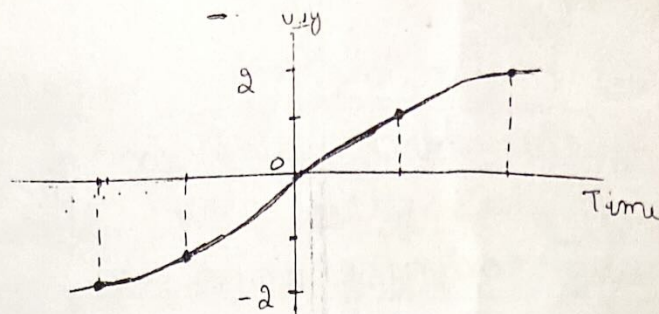


fig 2. (a) Analog waveform

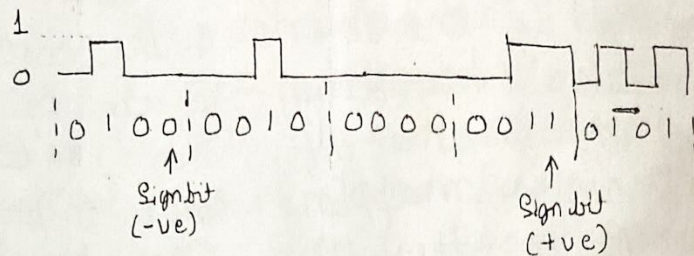


fig 2. (b) Digital representation.

The errors due to sampling & quantizing can be made so small that the difference b/w the analog & its digital reconstruction is not identified by a human observer.

Basic signal processing operations in digital communication.

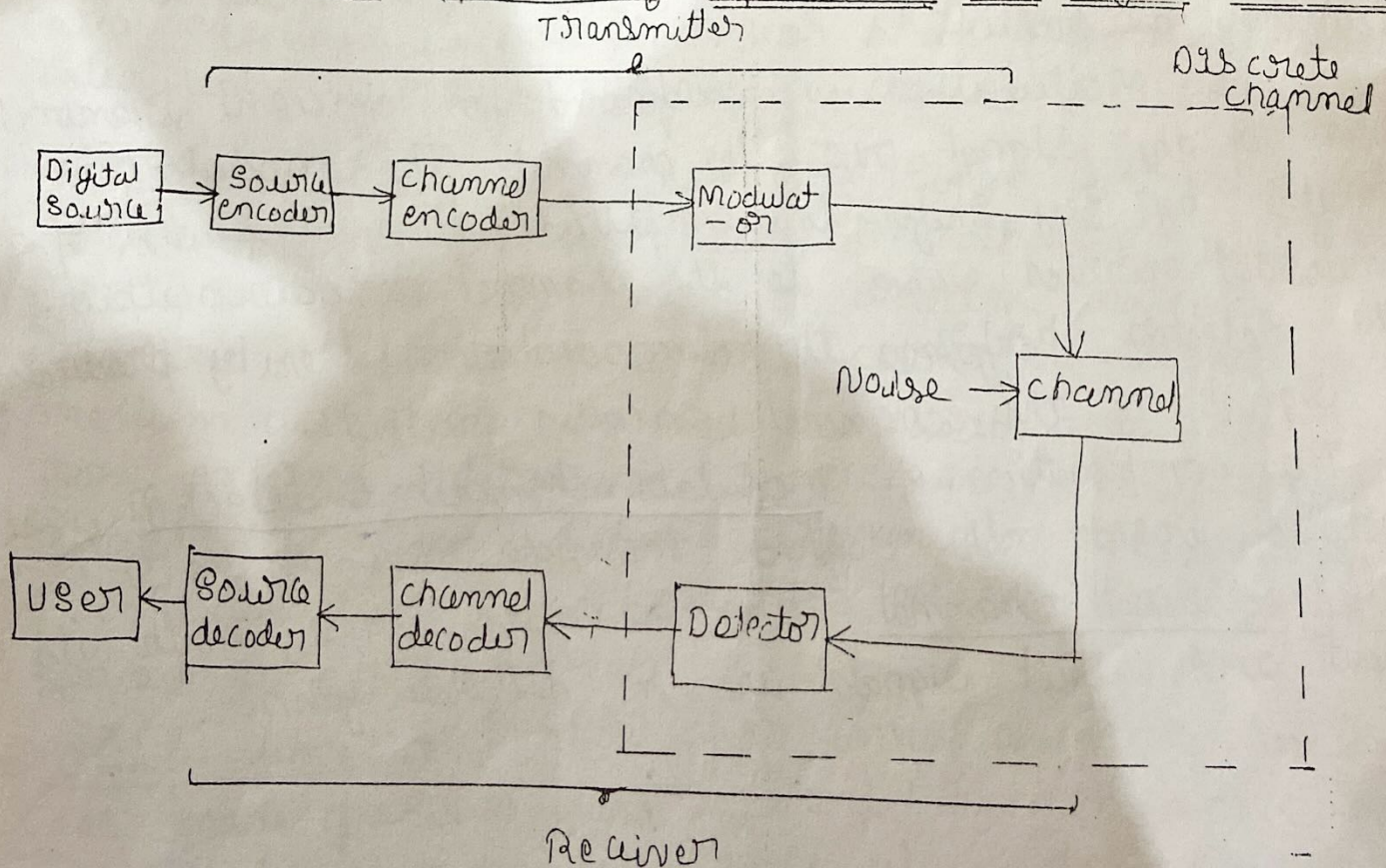


fig 3

Fig 3 shows the block diagram of digital communication system. It has three basic signal processing operations: source coding, channel coding and modulation. In source coding, the encoder maps the digital signal generated at the source o/p into another signal in digital form. The mapping is one to one & the objective is to eliminate or reduce redundancy so as to provide an efficient representation of the source o/p. The source decoder simply performs the inverse mapping and reproduce the original digital source o/p to the user. The benefit of source coding is a reduce bandwidth requirement.

The role of the channel encoder and channel decoder is to provide reliable communication over a noisy channel. This is satisfied by introducing redundancy in a prescribed fashion in the channel encoder & exploiting it in the decoder to reconstruct the original encoder input as accurately as possible.

Modulation is performed for efficient transmission of the signal over the channel. The modulation operates by key shifts in amplitude, freq or phase of a sinusoidal carrier wave to the channel encoder output.

The detector performs the demodulation, thereby producing a signal of the channel encoder output.

The combination of modulator, channel & detector, are enclosed inside the dashed rectangle shown in fig 3 is called discrete channel. it is so called since both its input and output signals are in discrete form.

Channel for digital communication's :-

The modulation and coding used in a digital comm system depend on the characteristics of the channel such as bandwidth, power, amplitude & phase response of the channel, and whether the channel is linear or non-linear and free from external interference.

The five specific channel issues are discussed below:

① Telephone channel: it is designed to provide voice-grade communication. Past 100 yrs it has evolved into worldwide network. The channel has a band-pass characteristic occupying the ~~freq~~ range 300 to 3400 Hz, a high SNR of about 30dB and approximately linear response. The transmission of voice signals results in a flat amplitude response over the pass-band of the channel, but phase response is not considered since ~~even~~ human ear is insensitive to phase-delay variations. but data & image transmission are strongly influenced by phase-delay variations, in such cases equalizer is used for linear phase response. Transmission rates upto 16.8 Kbps have been achieved over telephone lines.

② coaxial cable: It consists of a single wire conductor centered inside an outer conductor, which are insulated from each other by means of dielectric material. The primary advantage of co-axial cables are transmission medium are a relatively wide bandwidth and freedom from external interference. It requires closely spaced repeaters for long distance transmission. The data rate of co-axial cable is 274 megabits per second, with repeaters spaced at 1km intervals.

③ Optical fiber: An optical fiber consists of a very fine inner core made of silica glass, surrounded by a concentric layer called cladding that is also made up of glass. The glass in the core has a refractive index slightly higher than that of the glass in the cladding. A basic property of light

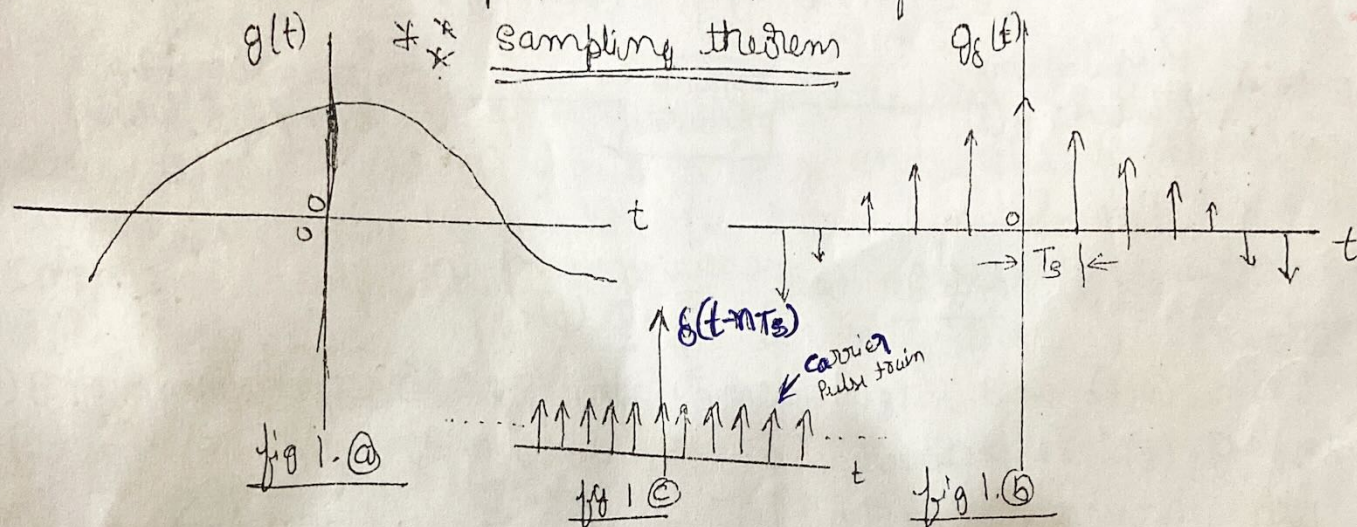
is that when a ray of light passes from a medium of high refractive index to another medium of low refractive index, the ray is bent back towards the medium with the higher RI. The difference b/w the RI of the core & cladding helps guide the propagation of the ray of light inside the core of the fibre from one end to the other. Compared to co-axial cables, optical fibers are smaller & they offer higher transmission bandwidths & longer repeater separation.

4) Microwave radio: It consists of transmitter and a receiver that are equipped with antennas as their own. The antennas are placed on towers at sufficient height to have the transmitter & receiver in line of sight of each other. The operating frequency ranges from 1 to 30 GHz. Under proper condition the radio channel operates as a non-dispersive transmission medium. During improper condition it undergoes multipath fading effect, means propagation b/w the transmitter & receiver takes place along several paths, thus the signal interfering with each other constructively or destructively.

5) Satellite channel: It consists of a satellite in geostationary orbit, an uplink from a ground station & a downlink to another ground station, they operate at microwave frequencies. Satellite channel may be viewed as a repeater in the sky, permitting communication over long distances at high bandwidth and relatively low cost.

3. Sampling process

Message signal may originate from a digital or analog source. If the message signal is derived from a digital source (e.g. digital computer), then for interaction it is in the right form for processing by a digital ~~computer~~ communication system. If the message signal is in analog form, as in speech signal or video signal, then it has to be converted into digital form. The sampling process is the first step performed in analog to digital conversion. In sampling process, a continuous-time signal is converted into a discrete-time signal by measuring the time signal at periodic instants of time.



Consider an analog signal $g(t)$ that is continuous in both time and amplitude. We assume that $g(t)$ is continuous has infinite duration but finite energy. A segment of the signal $g(t)$ is shown in fig 1(a).

Let the sample values of the signal $g(t)$ at times $t = 0, \pm T_s, \pm 2T_s, \dots$ be denoted by the series $\{g(nT_s), n = 0, \pm 1, \pm 2, \dots\}$ where T_s is sampling period and $f_s = \frac{1}{T_s}$ is sampling rate.

The discrete time signal $g_s(t)$, that results from the sampling process is given by

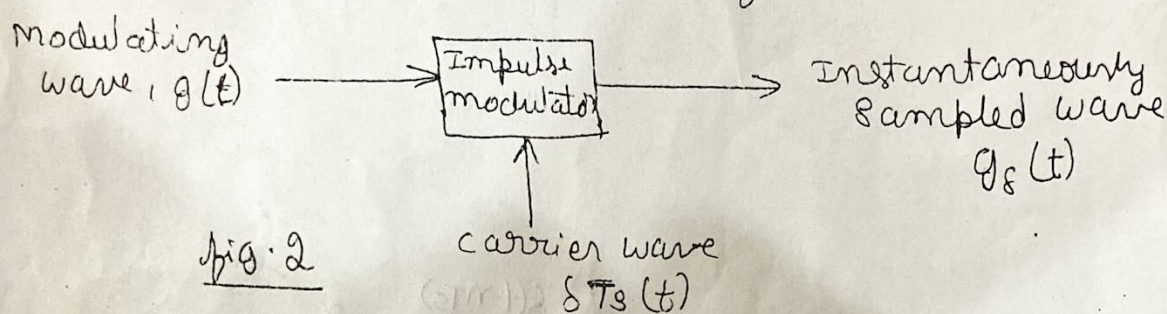
$$g_s(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \delta(t-nT_s) \quad \text{--- (1)}$$

where $\delta(t-nT_s)$ is a Dirac delta function located at time $t = nT_s$. In Eq (1) Each delta function in the series is weighted by the corresponding sample value of the input signal $g(t)$. fig 1(b) illustrates the construction of $g_s(t)$ from sample values of $g(t)$.

from the definition of a delta function, we have
 $g(nT_s) \delta(t - nT_s) = g(t) \delta(t - nT_s)$
 Hence Eq (1) becomes $g_s(t) = g(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$

$$= g(t) \delta_{T_s}(t) \quad \text{--- (2)}$$

where $\delta_{T_s}(t)$ is the Dirac comb or ideal sampling function. From Eq (2), we may view the discrete-time signal $g_s(t)$ as the o/p of an impulse modulator, which operates with $g(t)$ as the modulating wave and $\delta_{T_s}(t)$ as the carrier wave. This is depicted in fig. 2.



Let $G(f)$ and $G_s(f)$ denote the Fourier transforms of $g(t)$ and $g_s(t)$, respectively. For the Fourier transform of $\delta_{T_s}(t)$, we have

$$F[\delta_{T_s}(t)] = f_s \sum_{m=-\infty}^{+\infty} \delta(f - m f_s) \quad \text{--- (3)}$$

Transforming Eq (2) into frequency domain, we obtain

$$G_s(f) = G(f) * \left[f_s \sum_{m=-\infty}^{\infty} \delta(f - m f_s) \right] \quad \text{--- (4)}$$

where $*$ denotes convolution. Interchanging the order of summation and convolution yields.

$$G_s(f) = f_s \sum_{m=-\infty}^{\infty} G(f) * \delta(f - m f_s) \quad \text{--- (5)}$$

From the property of delta function, the convolution of $G(f)$ and $\delta(f - m f_s)$ equals $G(f - m f_s)$, Eq (5) is simplified as follows:

$$G_s(f) = f_s \sum_{m=-\infty}^{\infty} G(f - m f_s) \quad \text{--- (6)}$$

from Eq (6) we see that $G_s(f)$ represents a spectrum that is periodic in the frequency f with period $\frac{1}{T_s}$, but not necessarily continuous. In other words "The process of uniformly sampling a signal in the time domain results in a periodic spectrum in the freq domain with a period equal to the sampling rate". Thus $G_s(f)$ represents a periodic extension of the original spectrum $G(f)$. -

Now take Fourier transform on both sides of Eq (1) and noting that the Fourier transform of the delta function $\delta(t - nT_s)$ is equal to $\exp(-j2\pi n f T_s)$, we may thus write

$$G_s(f) = \sum_{n=-\infty}^{\infty} g(nT_s) \exp(-j2\pi n f T_s) \quad \text{--- (7)}$$

The relations that are derived, apply to any continuous-time signal $g(t)$ of finite energy and infinite duration. Suppose the signal is strictly band-limited, with no frequency components higher than W hertz. That is the Fourier transform $G(f)$ of the signal $g(t)$ has the property that $G(f)$ is zero for $f \geq W$. as shown in fig 3.a

If we choose the sampling period $T_s = 1/2W$, then

the corresponding spectrum $G_s(f)$ of the sampled signal $g_s(t)$ is as shown in figure 3.b.

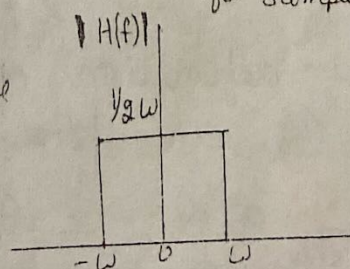
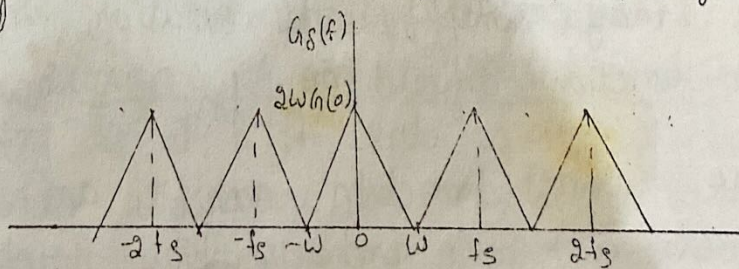
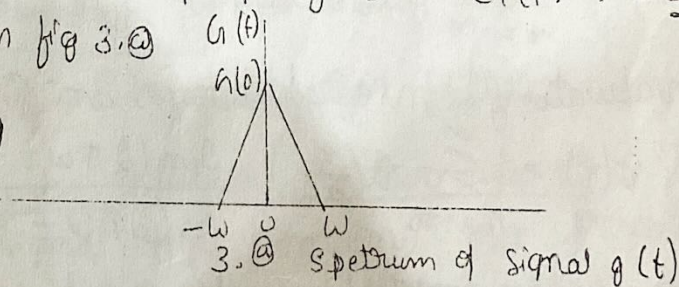
Putting $T_s = 1/2W$ in Eq (7) yields

$$G_s(f) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) \exp\left(-\frac{j\pi n f}{W}\right) \quad \text{--- (8)}$$

Putting $f_s = 2W$ in Eq (8) we have

$$G(f) = \frac{1}{2W} G_s(f) \quad -W < f < W$$

--- (9)



using Eq (8) we may also write

$$G(f) = \frac{1}{2W} \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) \exp\left(-\frac{j\pi n f}{W}\right) \quad -W < f < W \quad (10)$$

the sequence $g(n/2W)$ contains all the information of $g(t)$.
Reconstruction of signal $g(t)$ from the sequence of sample values $\{g(n/2W)\}$:-

Substitute Eq (10) in the formula for the inverse Fourier transform defining $g(t)$ in terms of $G(f)$, we get

$$\begin{aligned} g(t) &= \int_{-\infty}^{\infty} G(f) \exp(j2\pi ft) df \\ &= \int_{-W}^W \frac{1}{2W} \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) \exp\left(-\frac{j\pi n f}{W}\right) \exp(j2\pi ft) df \end{aligned}$$

interchanging the order of summation & integration:

$$g(t) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) \frac{1}{2W} \int_{-W}^W \exp\left[j2\pi f\left(t - \frac{n}{2W}\right)\right] df \quad (11)$$

evaluating integral terms in Eq (11) yields to

$$g(t) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) \frac{\sin(2\pi W t - n\pi)}{(2\pi W t - n\pi)} \quad (12)$$

We may simplify the notation in Eq (12) by using the sinc function, defined as

$$\text{sinc } x = \frac{\sin(\pi x)}{\pi x} \quad (13)$$

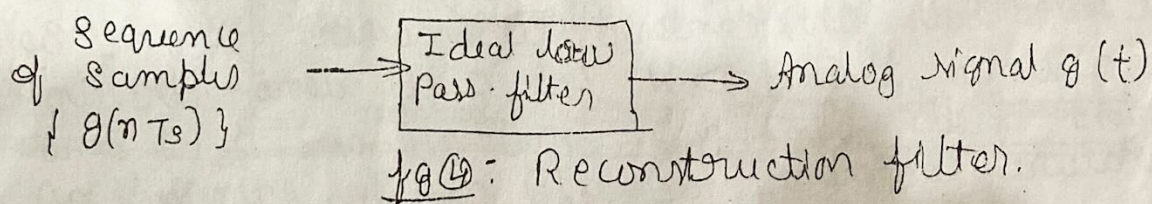
The sinc function exhibits an important property known as the interpolatory property, which is described as

$$\text{sinc } x = \begin{cases} 1 & x=0 \\ 0 & x=\pm 1, \pm 2, \dots \end{cases} \quad (14)$$

using the definition of the sinc function, we may rewrite Eq (12) as follows:

$$g(t) = \sum_{n=-\infty}^{\infty} g\left(\frac{n}{2W}\right) \text{sinc}(2Wt - n) \quad (15)$$

Eq (5) provides an interpolation formula for reconstructing the original signal $g(t)$ from the sequence of sample values $\{g(n/2W)\}$, with the sine function $\text{sinc}(2Wt)$. Each sample is multiplied by a delayed version of the interpolation function and all the resulting waveforms are added to obtain $g(t)$. Eq (5) also represents the response of an ideal low pass filter of Bandwidth W and zero transmission delay, which is produced by an input signal consisting of the sequence of samples $\{g(n/2W)\}$ for $-\infty \leq n \leq \infty$. By inspecting the spectrum of (5), we see that the original signal $g(t)$ may be recovered exactly from the sequence of samples $\{g(n/2W)\}$ by passing it through an ideal low pass filter of Bandwidth W . This is shown in fig (4). The ideal amplitude response of the reconstruction filter is shown in fig 3(c).



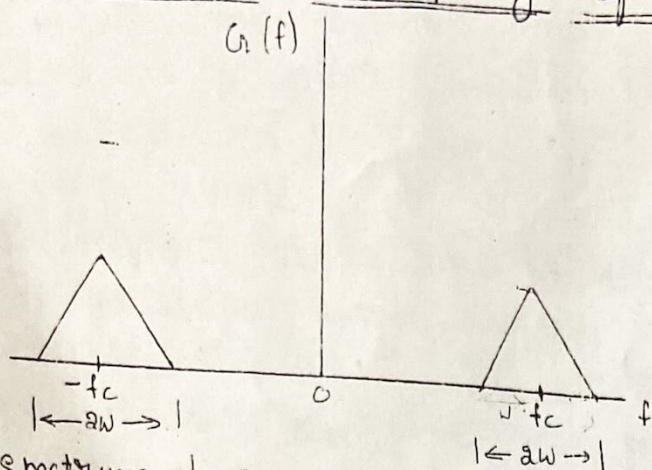
Sampling theorem can be stated in two parts:-

- ① If a finite energy signal $g(t)$ contains no frequencies higher than W hertz, it is completely determined by specifying its ordinates at a sequence of points spaced $1/2W$ seconds apart.
- ② If a finite energy signal $g(t)$ contains no frequencies higher than W hertz, it may be completely recovered from its ordinates at a sequence of points spaced $1/2W$ seconds apart.

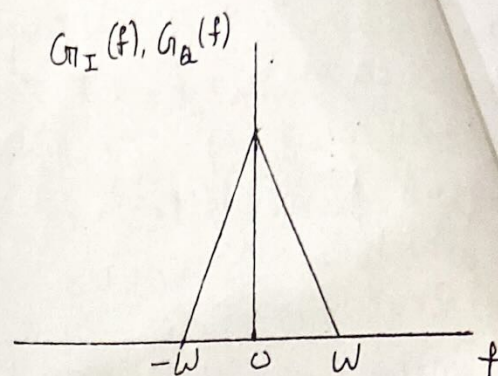
Part 1 of theorem is a restatement of Eq (6) and part 2 is the restatement of Eq (5).

The minimum sampling rate of $2W$ samples per second, for a signal bandwidth of W hertz is called Nyquist rate. $f_s = 2W$ correspondingly, the reciprocal, $1/2W$ is called Nyquist interval.

Quadrature Sampling of Band-pass Signal :



1. (a) Spectrum of Band-pass signal $g(t)$.



1. (b) Spectrum of low-pass in-phase component $g_I(t)$ & quadrature component $g_Q(t)$.

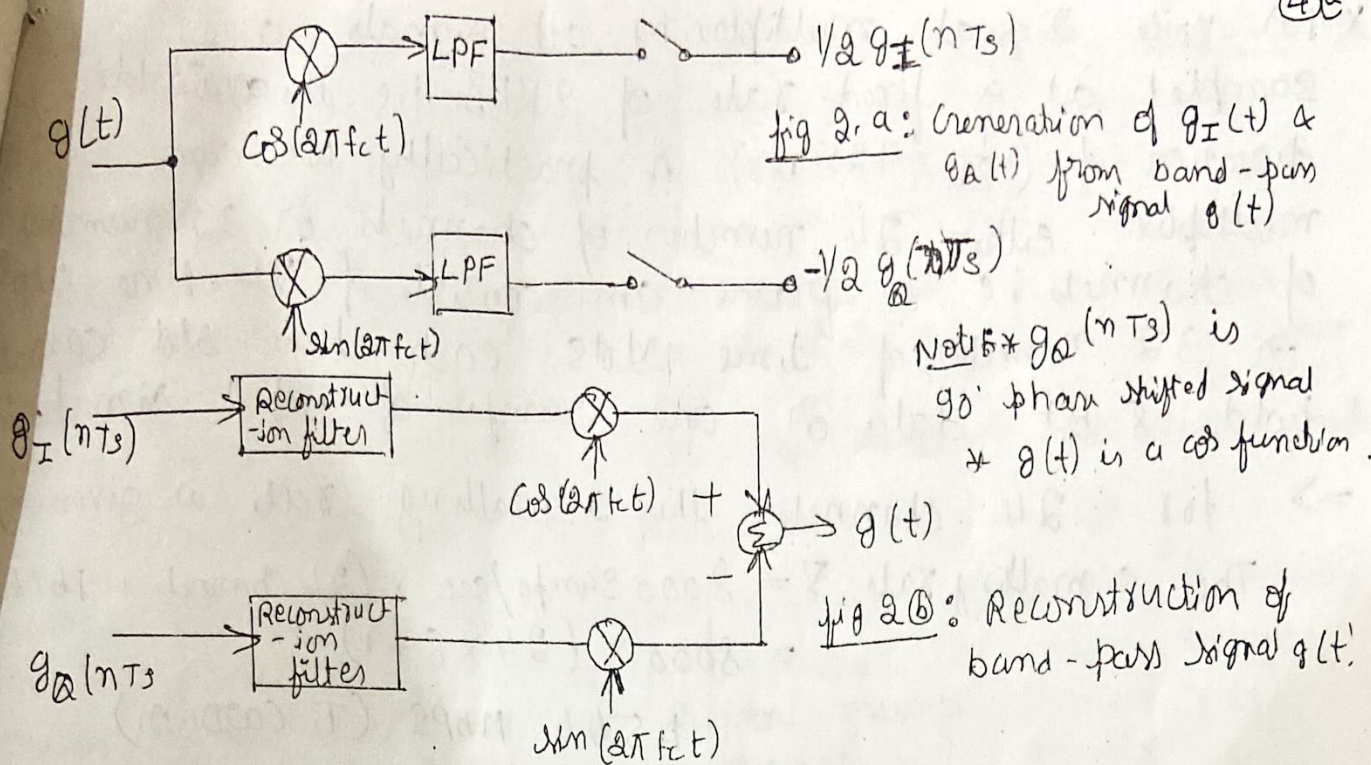
The band-pass signal can be represented in terms of its in-phase and quadrature components, each of which may be sampled separately.

Consider a band-pass signal $g(t)$ whose spectrum is limited to a bandwidth $2W$, centered around the freq f_c as in fig 1. (a). It is assumed that $f_c > W$. Let $g_I(t)$ denote the in-phase component of the band pass signal $g(t)$ and $g_Q(t)$ denote its quadrature component. So $g(t)$ can be expressed as

$$g(t) = g_I(t) \cos(2\pi f_c t) - g_Q(t) \sin(2\pi f_c t) \quad \text{--- (1)}$$

The $g_I(t)$ and $g_Q(t)$ may be obtained by multiplying the band pass signal $g(t)$ by $\cos(2\pi f_c t)$ and $\sin(2\pi f_c t)$ respectively and then suppressing the ~~same~~ frequency components by low-pass filters. Under the assumption that $f_c > W$, we find that $g_I(t)$ and $g_Q(t)$ are both low pass signals limited to $-W < f < W$ as in fig 1(b), each component may be sampled at the rate of $2W$ samples per second. This form of sampling is called quadrature sampling.

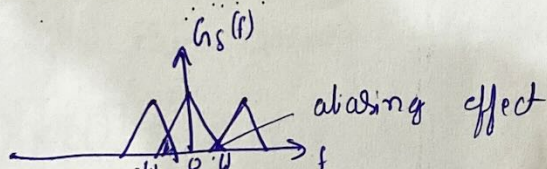
To reconstruct the original band-pass signal from its quadrature-sampled version, first reconstruct the in-phase component $g_I(t)$ and the quadrature component $g_Q(t)$ from their respective samples, multiply them respectively by $\cos(2\pi f_c t)$ and $\sin(2\pi f_c t)$ and then add the result.



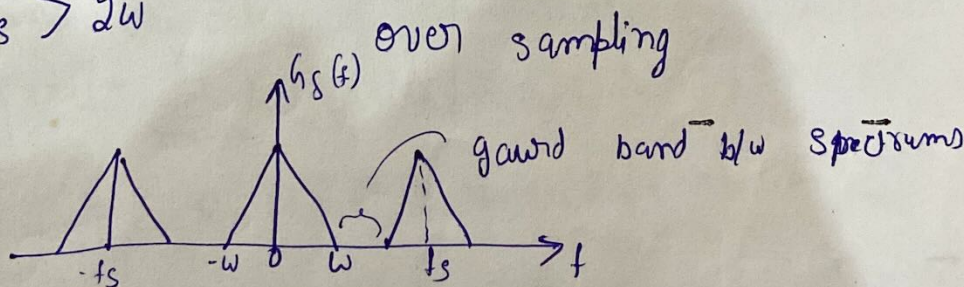
The operation that constitute quadrature sampling are shown in fig 2(a) and the reconstruction process are depicted in fig 2(b). Quadrature sampling is used for Band-pass signal

~~reconstruction of a message process from its samples.~~

(i) $f_s < 2W$ under sampling



(ii) $f_s > 2W$



* **sampling theorem:** A continuous time signal can be completely represented in its samples or recovered back if the sampling freq f_s is greater than or equal to the twice of highest freq component of the message signal f_m .

$$f_s \geq 2 f_m$$

* For voice signal multiplexing all signals are sampled at a fixed rate of 8 KHz i.e. the available duration of $(\frac{1}{8k} = 125 \mu s)$ & practically we can multiplex either 24 number of channels or 32 number of channels i.e. a frame can consists of 24 time slots or 32 number of time slots, each time slot can hold 8-bit data or one sample of input signal.

$\Rightarrow$ for 24 channels the signalling rate is given by
The signalling rate $\gamma = 8000 \text{ samples/sec} \times (24 \text{ channels} + 1 \text{ bit})$
 $= 8000 \times (24 \times 8 + 1)$
 $= 1.544 \text{ Mbps (T. carrier)}$

for 32 channels, the signalling rate γ
 $\gamma = 8000 \times (32 \times 8 + 1)$
 $= 2.048 \text{ Mbps (T. carrier)}$

The 1.544 Mbps & 2.048 Mbps are the two basic rates.

Reconstruction of a message process from its samples (4)

Consider stationary message process $X(t)$ with auto correlation function $R_X(\tau)$ and power spectral density $S_X(f)$. We assume that $S_X(f) = 0$ for $|f| \geq W$ — (1)

Suppose an infinite sequence of samples of the process taken at a uniform rate equal to $2W$, that is twice the highest frequency component of the process. Let $X'(t)$ be the reconstructed process, we may write ~~eq~~ based on interpolation formula as

$$X'(t) = \sum_{n=-\infty}^{\infty} X(n/2W) \text{sinc}(2Wt - n) \quad \text{--- (2)}$$

where $X(n/2W)$ is the random variable obtained by sampling message process $X(t)$ at time $t = n/2W$. The mean-square value of the error b/w the original message process $X(t)$ and the reconstructed message process $X'(t)$ equals

$$\begin{aligned} \mathcal{E} &= E[(X(t) - X'(t))^2] \\ &= E[X^2(t)] - 2E[X(t)X'(t)] + E[(X'(t))^2] \quad \text{--- (3)} \end{aligned}$$

By taking the first expectation term on the right side of eq (3) as the mean-square value of $X(t)$, which equals $R_X(0)$;

Thus $E[X^2(t)] = R_X(0) \quad \text{--- (4)}$

By taking the second expectation term in eq (3)

$$E[X(t)X'(t)] = E\left[X(t) \sum_{n=-\infty}^{\infty} X(n/2W) \text{sinc}(2Wt - n)\right]$$

(Interchanging the order of summation & expectation)

$$\begin{aligned} E[X(t)X'(t)] &= \sum_{n=-\infty}^{\infty} E[X(t)X(n/2W)] \text{sinc}(2Wt - n) \\ &= \sum_{n=-\infty}^{\infty} R_X(t - \frac{n}{2W}) \text{sinc}(2Wt - n) \quad \text{--- (5)} \end{aligned}$$

For a stationary process, the expectation $E[X(t)X'(t)]$ is independent of time t . Hence putting $t=0$ in the right side of eq (5) and recognizing that

$$R_X(-\frac{n}{2W}) = R_X(\frac{n}{2W})$$

~~Aperture effect -~~

~~Distortion caused by lengthening of samples.~~

we can write

$$E[X(t)X'(t)] = \sum_{n=-\infty}^{\infty} R_X(n/2\omega) \text{sinc}(-n) \quad \text{--- (6)}$$

we may write $R_X(\tau)$ in terms of its samples taken at $\tau = n/2\omega$ using interpolation formula as

$$R_X(\tau) = \sum_{n=-\infty}^{\infty} R_X(n/2\omega) \text{sinc}(2\omega\tau - n) \quad \text{--- (7)}$$

from Eq (6) & (7) we can write

$$E[X(t)X'(t)] = R_X(0) \quad \text{--- (8)}$$

The ~~last~~ third expectation term on the right side of Eq (3), again using Eq (2) we can write

$$\begin{aligned} E[(X'(t))^2] &= E\left[\sum_{n=-\infty}^{\infty} X(n/2\omega) \text{sinc}(2\omega t - n) \sum_{k=-\infty}^{\infty} X(k/2\omega) \text{sinc}(2\omega t - k)\right] \\ &= E\left[\sum_{n=-\infty}^{\infty} \text{sinc}(2\omega t - n) \sum_{k=-\infty}^{\infty} X(n/2\omega) X(k/2\omega) \text{sinc}(2\omega t - k)\right] \end{aligned}$$

interchanging the order of expectation & inner summation

$$\begin{aligned} E[(X'(t))^2] &= \sum_{n=-\infty}^{\infty} \text{sinc}(2\omega t - n) \sum_{k=-\infty}^{\infty} E[X(n/2\omega) X(k/2\omega)] \text{sinc}(2\omega t - k) \\ &= \sum_{n=-\infty}^{\infty} \text{sinc}(2\omega t - n) \sum_{k=-\infty}^{\infty} R_X\left(\frac{n-k}{2\omega}\right) \text{sinc}(2\omega t - k) \quad \text{--- (9)} \end{aligned}$$

By simplifying Eq (9) in accordance to Eq (7) we obtain

$$E[(X'(t))^2] = \sum_{n=-\infty}^{\infty} R_X\left(t - \frac{n}{2\omega}\right) \text{sinc}(2\omega t - n)$$

$$E[(X'(t))^2] = R_X(0) \quad \text{--- (10)}$$

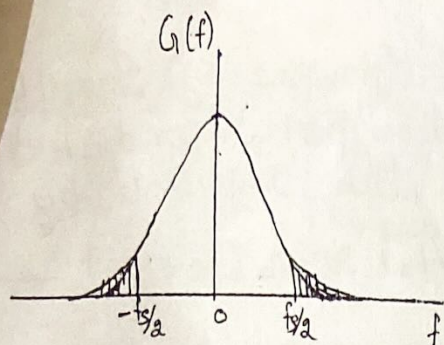
Substituting Eq 4, 8 & 10 into Eq (3) we get the result

$$\mathcal{E}_0 = 0$$

Therefore the sampling theorem for message processes can be stated as: "If a stationary message process contains no frequencies higher than ω hertz, it may be reconstructed from its samples at a sequence of points spaced $1/2\omega$ seconds apart with zero mean squared error."

~~not to be~~

Signal distortion in sampling:-



1(a) - Spectrum of finite energy signal $g(t)$ whose spectrum decreases with increase in freq.

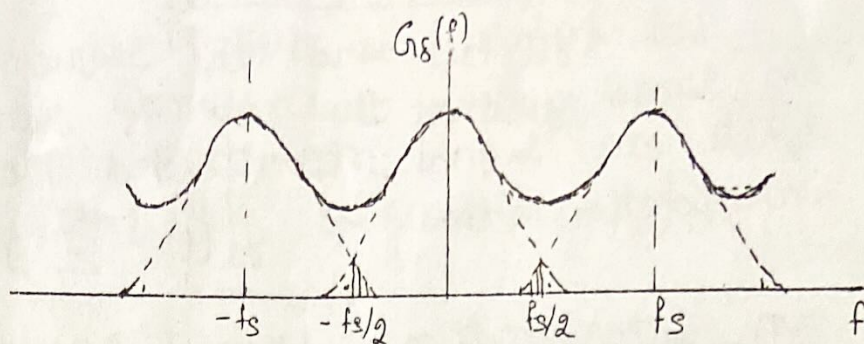


fig 1(b)

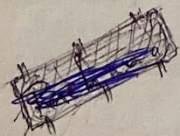
consider a signal $g(t)$ whose spectrum $G(f)$ decreases with increasing freq f without limit, as shown in fig 1(a). The spectrum $G_s(f)$ of the discrete time signal $g_s(t)$, resulting from the use of idealized sampling, is the sum of $G(f)$ and an infinite number of freq-shifted replicas of it, in accordance to

$$Eq \quad G_s(f) = f_s \sum_{m=-\infty}^{\infty} G(f - m f_s). \quad \text{The replicas of } G(f) \text{ are}$$

shifted in freq by multiples of the sampling rate f_s . Two replicas of $G(f)$ are shown in fig 1(b), one at f_s and the other at $-f_s$. The use of a low pass reconstruction filter, with its pass band extending from $-f_s/2$ to $f_s/2$ no longer yields an undistorted original signal $g(t)$. We find that portions of the freq-shifted replicas are folded over inside the desired spectrum, high frequencies in $G(f)$ are reflected into low frequencies in $G_s(f)$, as shown in fig 1(a) & 1(b).

This phenomenon is called aliasing or foldover. When this effect occurs, information is inevitably lost in the sampling process.

The effect of aliasing on the output of the reconstruction filter depends on both the amplitude and phase components of the original spectrum $G(f)$, making an exact analysis of the effect difficult.

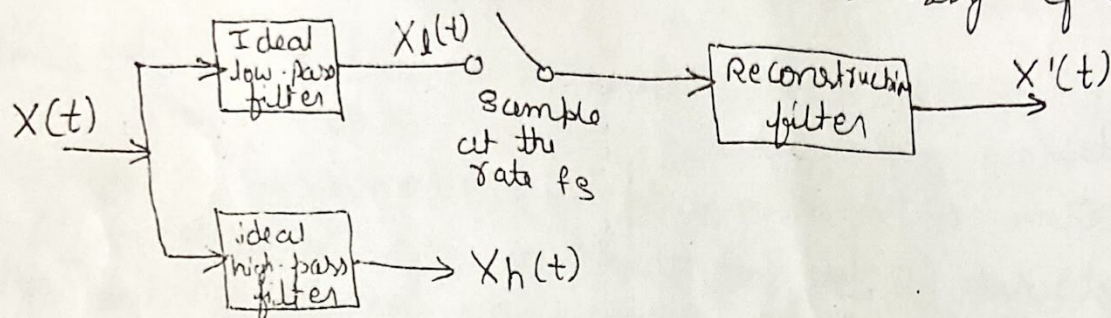


Signal to distortion ratio

consider a message process $X(t)$ that is wide-sense stationary but not strictly band-limited. for a sampling rate f_s , we have

$$S_X(f) \neq 0 \quad |f| > f_s/2 \quad \text{--- (1)}$$

where $S_X(f)$ is the power spectral density of the process.



The message process $X(t)$ is passed through an ideal low-pass filter with transfer function

$$H_L(f) = \begin{cases} 1 & |f| < W \\ 0 & |f| > W \end{cases} \quad \text{--- (2)}$$

Note: $H(f) = \frac{o/p}{s/p}$

Let $X_L(t)$ denote the random process resulting at the filter output, which is strictly band limited to $f_s/2$. that is

$$S_{X_L}(f) = \begin{cases} S_X(f) & |f| < f_s/2 \\ 0 & |f| \geq f_s/2 \end{cases} \quad \text{--- (3)}$$

where $S_{X_L}(f)$ is the power spectral density of $X_L(f)$.

In accordance with sampling theorem, the process $X_L(t)$ may be reconstructed from its samples exactly. Let $X'(t)$ denote the process reconstructed from samples of $X_L(t)$. But $X'(t)$ is different from the original process $X(t)$, since $X_L(t)$ is different from $X(t)$.

To evaluate the average power of this difference, let the process $X(t)$ be passed through an ideal high pass filter as shown in lower path of fig. This second filter has the transfer function

$$H_H(f) = \begin{cases} 0 & |f| < f_s/2 \\ 1 & |f| > f_s/2 \end{cases} \quad \text{--- (4)}$$

Let $X_h(t)$ denote the random process resulting at the high-pass filter o/p. Its power spectral density is defined by

$$S_{X_h}(f) = \begin{cases} 0 & |f| \leq f_s/2 \\ S_X(f) & |f| > f_s/2 \end{cases} \quad \text{--- (5)}$$

Since $H_l(f) + H_h(f) = 1$ for all f --- (6)

it follows that $X_l(t) + X_h(t) = X(t)$ for all t ---

or $X_h(t) = X(t) - X_l(t)$ for all t --- (7)

we may view $X_h(t)$ as the error b/w the process $X(t)$ and the process $X_l(t)$ reconstructed from samples $X_l(t)$. Therefore mean squared error can be expressed as

$$\mathcal{E} = E[(X_l(t) - X(t))^2]$$

$$= \int_{-\infty}^{\infty} S_{X_h}(f) df$$

$$= \int_{|f| > f_s/2} S_X(f) df \quad \text{--- (8)}$$

The total power of the process $X(t)$ is given by

$$P = \int_{-\infty}^{\infty} S_X(f) df \quad \text{--- (9)}$$

Hence signal to distortion ratio (SDR) as

$$\begin{aligned} \text{SDR} &= P/\mathcal{E} \\ &= \frac{\int_{-\infty}^{\infty} S_X(f) df}{\int_{|f| > f_s/2} S_X(f) df} \quad \text{--- (10)} \end{aligned}$$

from Eq (10), we can note that SDR increases as the sampling rate f_s is increased.

Practical aspects of sampling and signal recovery

▷ Ordinary samples of finite duration:- @ Natural Sampling

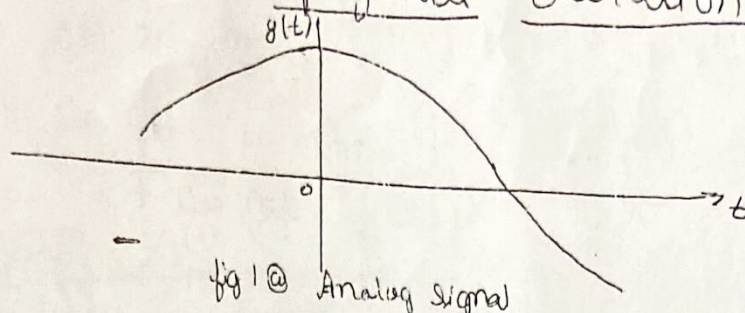


fig 1(a) Analog signal

infinite rectangular pulses

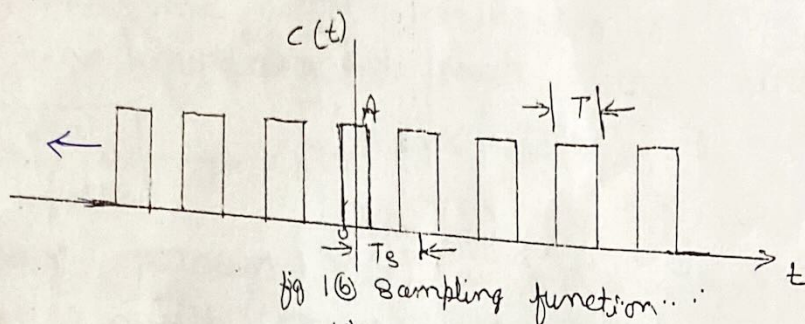


fig 1(b) Sampling function...

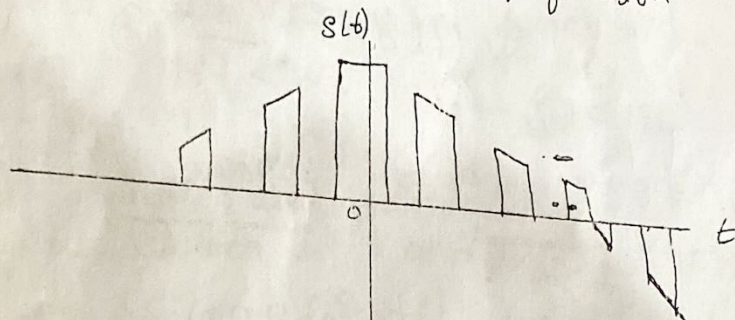


fig 1(c) Sampled signal

Let an arbitrary analog signal $g(t)$ be applied to a switching ckt controlled by a sampling function $c(t)$ that consists of an infinite succession of rectangular pulses of amplitude A , duration T and occurring with period T_s . The o/p of the switching ckt is denoted by $s(t)$. The waveforms $g(t)$, $c(t)$ and $s(t)$ are illustrated in parts (a), (b) & (c) in fig 1. We see that the switching operation extracts from the analog signal $g(t)$ successive portions of predetermined duration T , taken regularly at the rate $f_s = 1/T_s$. The sampled signal $s(t)$ consists of a sequence of positive & negative pulses as in fig 1(c). We have

$$s(t) = c(t) \cdot g(t) \quad \text{--- (1)}$$

$c(t)$ may be expressed in the form of a complex Fourier series as (1)

$$c(t) = f_s T A \sum_{n=-\infty}^{\infty} \text{sinc}(n f_s T) \exp(j 2 \pi n f_s t) \quad \text{--- (2)}$$

Substituting Eq (2) in (1) we get

$$s(t) = f_s T A \sum_{n=-\infty}^{\infty} \text{sinc}(n f_s T) \exp(j 2 \pi n f_s t) g(t) \quad \text{--- (3)}$$

Taking Fourier transform on both the sides of Eq (3) using frequency shifting property of the Fourier transform we get

$$S(f) = f_s T A \sum_{m=-\infty}^{\infty} \text{sinc}(m f_s T) G(f - m f_s) \quad \text{--- (4)}$$

where $S(f) = F[s(t)]$ and $G(f) = F[g(t)]$.

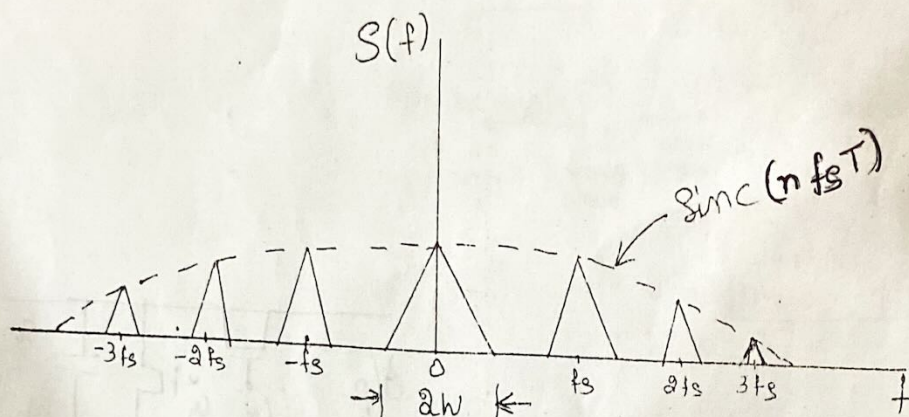


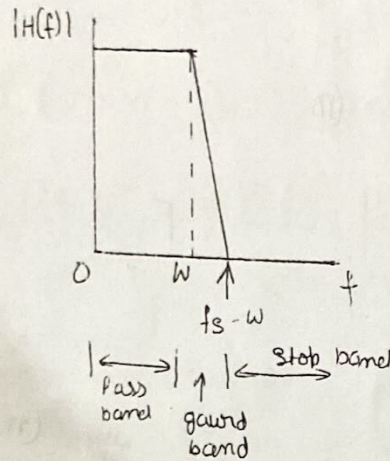
fig (2)

The relationship b/w the spectrum $G(f)$ and $S(f)$ is shown in fig (2). Assuming that $g(t)$ contains no freq lying outside the band $-W$ to W and assuming that sampling rate f_s is greater than $2W$, so there is no aliasing. The original signal $g(t)$ can be recovered from $s(t)$ with no distortion by passing $s(t)$ through an ideal low-pass filter whose bandwidth B , as before, satisfies the condition $W < B < f_s - W$, therefore the use of sampling pulses of finite duration has no important effects on the sampling process.

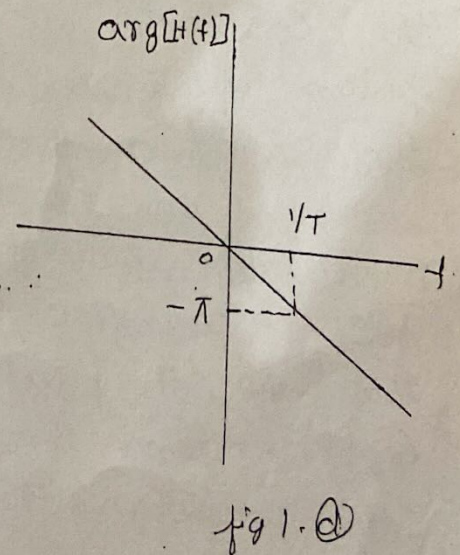
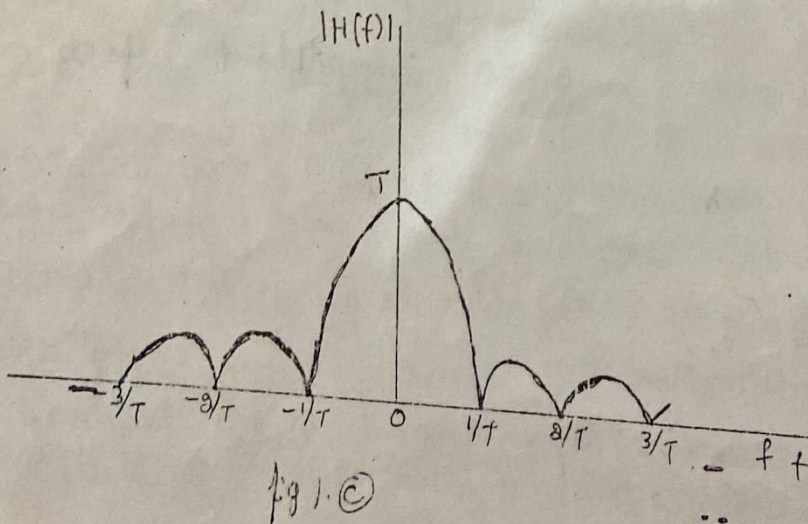
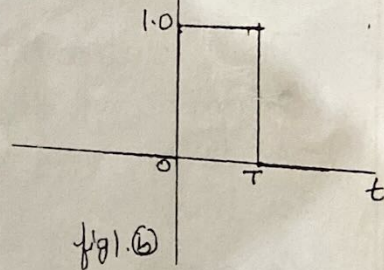
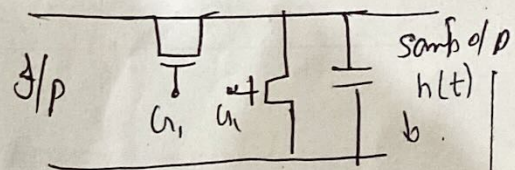
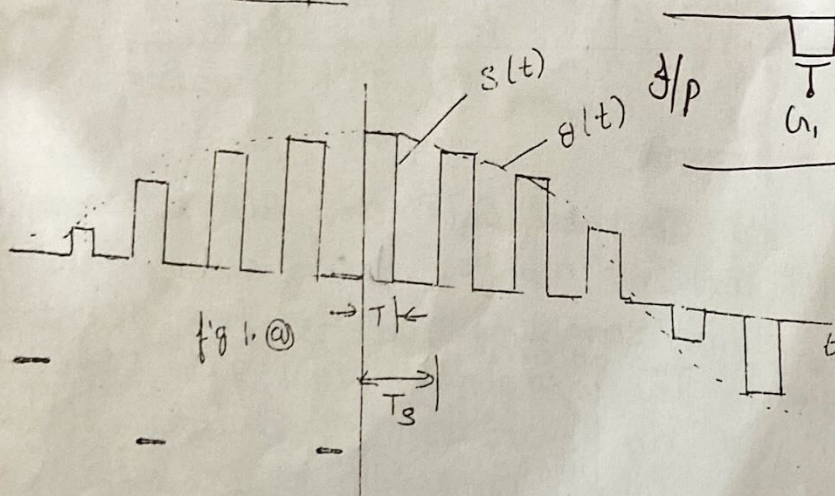
consider the case when $T.A = 1$, so that the each rectangular pulse of the sampling function $c(t)$ has unit area. Then comparing Eq (4) with $G_s(f) = f_s \sum_{m=-\infty}^{\infty} G(f - m f_s)$, we see that as the pulse duration T approaches zero, $S(f)$ approaches $G_s(f)$, means the discrete-time signal $g_s(t)$ produced by ideal sampling represents the limiting form of the sampled signal

$s(t)$ as T approaches zero, with $TA = 1$.

Note If the sampling rate is higher than Nyquist then it is easier to design low pass filter to recover original signal. With such a sampling rate, there are each of width $f_s - 2W$ low freq shifted replicas of $G(f)$

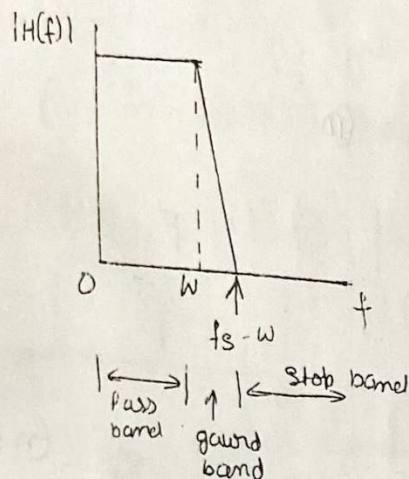


2) Flat-top samples

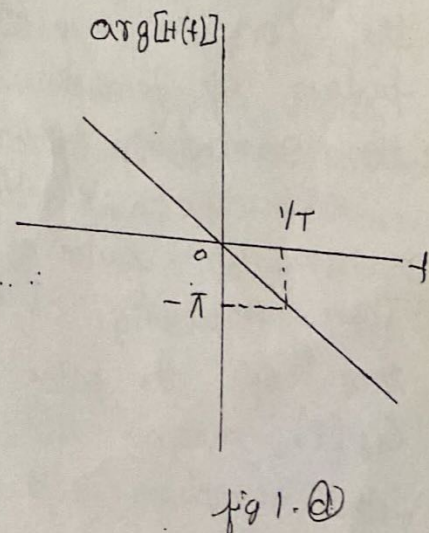
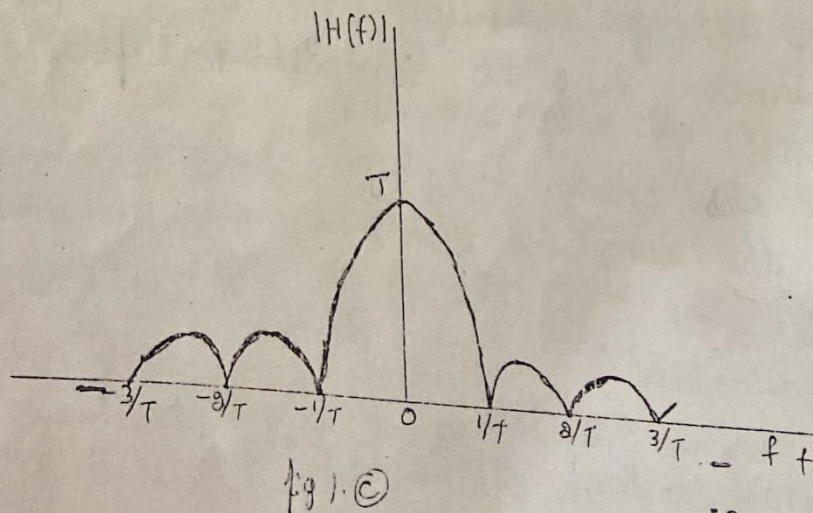
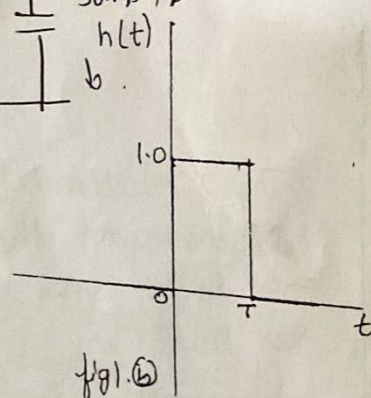
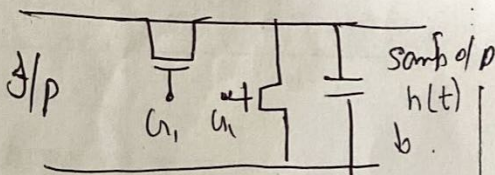
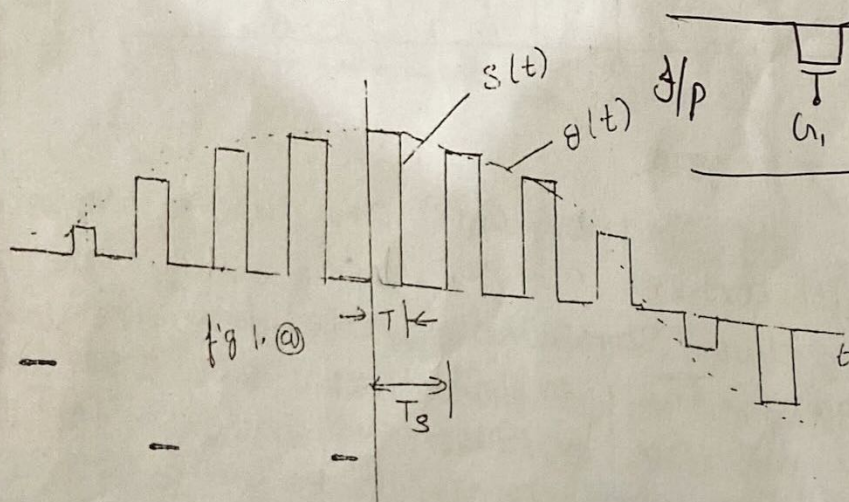


$s(t)$ as T approaches zero, with $TA = 1$.

Note If the sampling rate is higher than Nyquist rate, then it is easier to design low pass filter to recover the original signal. With such a sampling rate, there are gaps each of width $f_s - 2W$ b/w freq shifted replicas of $G(f)$



2) Flat-top samples



consider an analog signal $g(t)$ is sampled instantaneously at the rate $f_s = 1/T_s$ and that the duration of each sample is lengthened to T as shown in fig 1. (a). (3)

Using $s(t)$ to denote the sequence of flat-top pulses generated in this way, we may write

$$s(t) = \sum_{n=-\infty}^{\infty} g(nT_s) h(t - nT_s) \quad \text{--- (1) } \checkmark$$

The $h(t)$ is a rectangular pulse of unit amplitude and duration T as in fig 1 (b), can be defined as

$$h(t) = \begin{cases} 1 & 0 < t < T \\ 0 & \dots t < 0 \text{ and } t > T \end{cases}$$

$$= \text{rect}\left(\frac{t}{T} - \frac{1}{2}\right) \quad \text{--- (2)}$$

The discrete time signal $g_s(t)$ obtained by sampling $g(t)$ is given by

$$g_s(t) = \sum_{n=-\infty}^{\infty} g(nT_s) \delta(t - nT_s) \quad \text{--- (3)}$$

convolving $g_s(t)$ with $h(t)$ we get

$$g_s(t) * h(t) = \int_{-\infty}^{\infty} g_s(\tau) h(t - \tau) d\tau$$

$$= \int_{-\infty}^{\infty} \sum_{n=-\infty}^{\infty} g(nT_s) \delta(\tau - nT_s) h(t - \tau) d\tau$$

$$= \sum_{n=-\infty}^{\infty} g(nT_s) \int_{-\infty}^{\infty} \delta(\tau - nT_s) h(t - \tau) d\tau \quad \text{--- (4)}$$

from the ~~sift~~ ^{sifting} property of the delta function, we have

$$g_s(t) * h(t) = \sum_{n=-\infty}^{\infty} g(nT_s) h(t - nT_s) \quad \text{--- (5)}$$

Therefore from eq (4) & (5) it follows that $s(t)$ is equivalent to the convolution of $g_s(t)$ and pulse $h(t)$

$$s(t) = g_s(t) * h(t) \quad \text{--- (6)}$$

Taking Fourier transform on both sides of eq (6).

$$S(f) = G_s(f) H(f) \quad \text{--- (7)}$$

$$\text{W.K.T. } G_s(f) = f_s \sum_{m=-\infty}^{\infty} G(f - m f_s) \quad \text{--- (8)}$$

Subst eq (8) in eq (7)

$$S(f) = f_s \sum_{m=-\infty}^{\infty} G(f - m f_s) H(f) \quad \text{--- (9)}$$

from fig 1.© we see that by using flat-top samples we have introduced amplitude distortion as well as a delay of $T/2$. The distortion caused by lengthening the samples is referred to as the aperture effect.

This distortion may be corrected by connecting an equalizer in cascade with the low-pass reconstruction filter. The equalizer has the effect of decreasing the in-band loss of the reconstruction filter as the frequency increases in such a manner as to compensate for the aperture effect.

from eq (2) we find that

$$H(f) = T \text{sinc}(fT) \exp(-j\pi fT) \quad \text{--- (1a)}$$

which is plotted in fig 1.©.

The amplitude response of the equalizer is given by

$$\frac{1}{|H(f)|} = \frac{1}{T \text{sinc}(fT)} = \frac{1}{T} \frac{\pi f T}{\sin(\pi f T)}$$

The amount of equalization needed is usually small.

3) Sample and hold ckt for signal recovery:

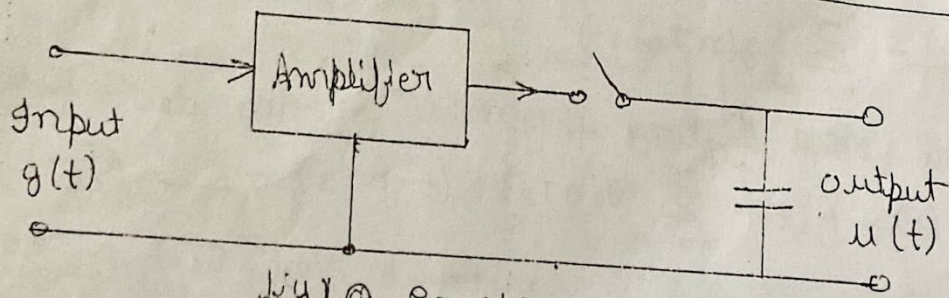


fig 1© sample & hold ckt

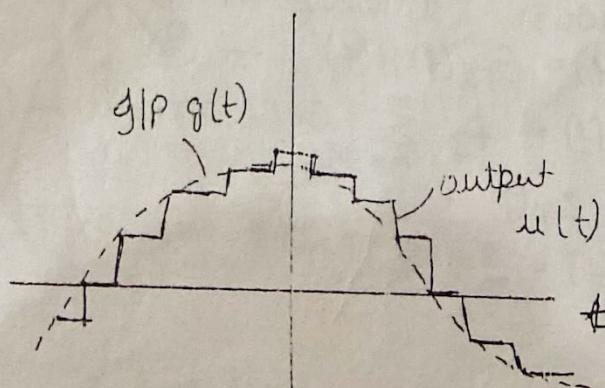


fig 1© idealized output waveform of the ckt.

In both natural sampling and flat-top sampling the spectrum of the sampled signal is scaled by the ratio T/T_s where T is sampling pulse duration and T_s is the sampling period. This ratio is quite small, with the result that the signal power at the output of the low-pass reconstruction filter in the receiver is small. So amplification is needed. This is done by sample & hold ckt as in fig 10. consists of an amplifier of unity gain and low o/p impedance, a switch and a capacitor. The switch is timed to close only for a small duration T of each sampling pulse, during which time the capacitor charges up to a v_g level equal to that of the input sequence. When the switch is open, the capacitor retains its v_g level until the next closure of the switch. Thus the sample and hold circuit, in its ideal form, produces an output waveform that represents a staircase interpolation of the original analog signal as shown in fig 10.

The o/p of the sample & hold ckt is defined by

$$u(t) = \sum_{n=-\infty}^{\infty} g(nT_s) h(t - nT_s) \quad \text{--- (1)}$$

where $h(t)$ is an impulse response representing the action of the sample & hold ckt: that is

$$h(t) = \begin{cases} 1 & 0 < t < T_s \\ 0 & t < 0 \text{ \& } t > T_s \end{cases} \quad \text{--- (2)}$$

The spectrum of the o/p is given by

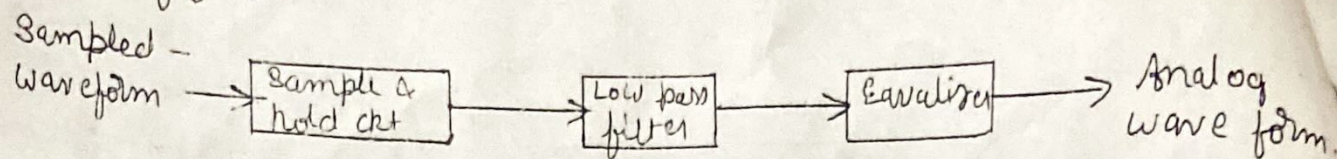
$$U(f) = T_s \sum_{m=-\infty}^{\infty} H(f) G(f - mT_s) \quad \text{--- (3)}$$

The transfer function $H(f)$ is given by

$$H(f) = T_s \text{sinc}(fT_s) \exp(-j\pi fT_s) \quad \text{--- (4)}$$

To recover the original signal $g(t)$ without distortion, we need to pass the o/p of the sample and hold circuit through a low pass filter designed to remove components of the spectrum $U(f)$ at multiples of the

sampling rate f_s and an equaliser whose amplitude response equals $1/|H(f)|$. These operations are shown in fig 8.



Pulse-Amplitude modulation (PAM):

In PAM, the amplitude of a carrier consisting of a periodic train of rectangular pulses is varied in proportion to sample values of a message signal. In this type of modulation, the pulse duration is held constant, but the amplitude of each rectangular pulse is varied as the value of the message signal at the leading edge of the pulse, so that PAM is exactly same as flat top sampling. The waveform is same as shown in flat top sampling.

Let $v(t)$ denote the pulse of carrier. Then the PAM wave, $s(t)$ is given by

$$s(t) = \sum_{n=-\infty}^{\infty} g(nT_s) v(t - nT_s) \quad \text{--- (1)}$$

where $g(nT_s)$ are sample values of the message signal $g(t)$ and T_s is the sampling period. Thus the definition of PAM given in terms of rectangular pulses may be viewed as a special case of Eq (1) by setting $v(t) = h(t)$, where $h(t)$ is defined by

$$h(t) = \begin{cases} 1 & 0 < t < T \\ 0 & t < 0 \text{ and } t > T \end{cases}$$

$$= \text{rect} \left(\frac{t}{T} - \frac{1}{2} \right) \quad \text{--- (2)}$$

Transmission bandwidth required can be satisfied by finding a suitable pulse shape for $v(t)$.

Note 1 PDM - pulse amplitude is kept constant, but pulse duration is varied in proportion to sample values of message signal.

Time-division multiplexing

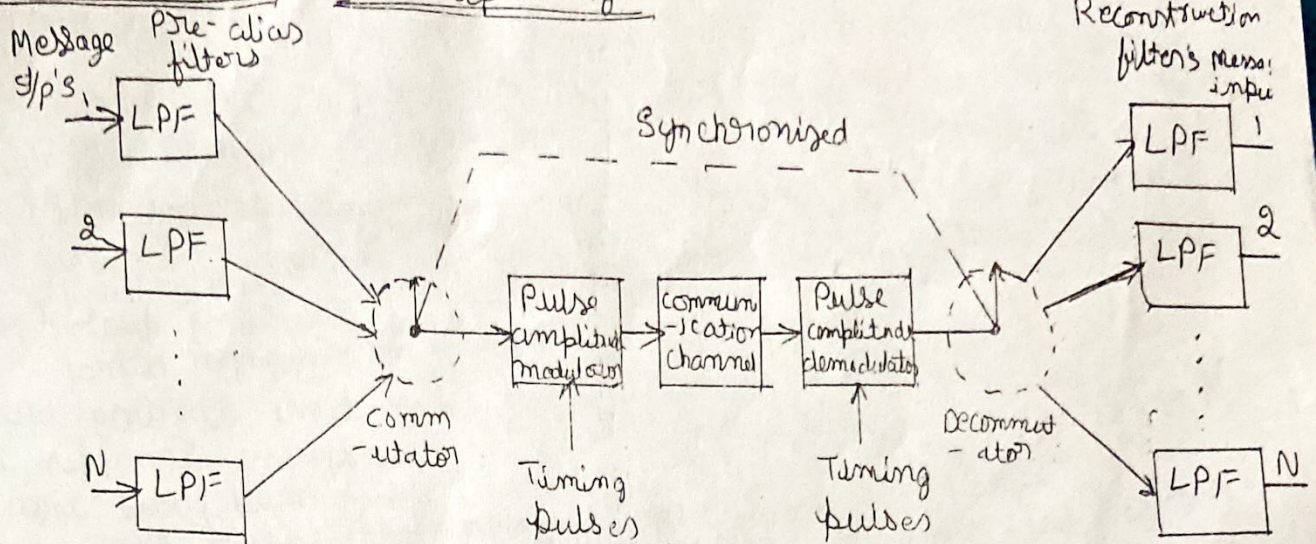


Fig 1

An important feature of pulse-amplitude modulation is a conservation of time. For a given message signal, transmission of the associated PAM wave engages the communication channel for only a fraction of the sampling interval on a periodic basis. Hence some of the time interval b/w adjacent pulses of the PAM wave is cleared for use by other independent message signal on a time shared basis. By doing this we obtain TDM, which enables the joint utilization of a common channel by the independent message signals without mutual interference.

The concept of TDM is shown in Fig 1. Each input message signal is first restricted in bandwidth by a low-pass pre-alias filter to remove the frequencies ~~in first restricted~~ ~~in bandwidth~~ that are non-essential to an adequate signal representation. The pre-alias filter outputs are then applied to a commutator, which is implemented using electronic switching circuitry. The function of the commutator is two-fold: ① To take a narrow sample of each of the N input messages at a rate f_s that is slightly higher than $2W$ where W is the cutoff freq of the pre-alias filter and ② to sequentially interleave these N samples inside a sampling interval $T_s = 1/f_s$. After the commutation process, the multiplexed signal is applied to a pulse-amplitude modulator.

to transform the multiplexed signal into a form suitable for transmission over the communication channel.

Suppose the N message signals to be multiplexed have similar spectral properties, then sampling rate for each message signal is determined using sampling theorem. then we have

$$T_x = \frac{T_s}{N}$$

$T_s \rightarrow$ sampling period of each message signal

$T_x \rightarrow$ time spacing b/w adjacent samples in the time-multiplexed signal.

Hence, the use of TDM introduces a bandwidth expansion factor N , because the scheme must squeeze N samples drawn from N independent message signals into a time slot equal to one sampling interval.

At the receiving end of the system, the received signal is applied to a pulse amplitude demodulator, which performs reverse operation of the pulse amplitude modulator. The short pulses produced at the pulse demodulator o/p are distributed to the appropriate low pass reconstruction filters by means of a decommutator, which operates in synchronism with the commutator in the transmitter.

The TDM system is highly sensitive to dispersion in the common channel, so accurate equalization of both the amplitude and pulse responses of the channel is necessary to ensure a satisfactory operation of the system. by "equalization" means the process of correcting channel induced dispersion.

equalized : process of correcting channel induced dispersion

Sampling process

S18

①. Find the Nyquist rate and Nyquist interval for following signals.

i). $m(t) = \frac{1}{2\pi} \cos(4000\pi t) \cdot \cos(1000\pi t)$

ii). $m(t) = \frac{\sin 500\pi t}{\pi t}$

i). $m(t) = \frac{1}{2\pi} \cdot \cos(4000\pi t) \cdot \cos(1000\pi t)$

$$= \frac{1}{2\pi} \times \frac{2}{2} (\cos(4000\pi t + 1000\pi t) + \cos(4000\pi t - 1000\pi t))$$

$$= \frac{1}{4\pi} \{ \cos 3000\pi t + \cos 5000\pi t \}$$

$$= \frac{1}{4\pi} \{ \cos 2\pi f_1 t + \cos 2\pi f_2 t \}$$

$$f_1 = 1500, \quad f_2 = 2500 \text{ Hz}$$

$$\text{Highest frequency Component} = f_2 = 2500 \text{ Hz}$$

$$\text{NR} = \text{Nyquist rate} = 2 \times 2500 \\ = 5000 \text{ Hz}$$

$$\text{Nyquist interval} = \frac{1}{\text{NR}} = \frac{1}{5000} = 0.2 \text{ msec}$$

ii). $m(t) = \frac{\sin(500\pi t)}{\pi t} = \frac{\sin(2\pi f_1 t)}{\pi t}$

$$f_1 = \frac{500}{2} = 250 \text{ Hz}$$

$$\text{NR} = \text{Nyquist rate} = 2 \times 250 = 500 \text{ Hz}$$

$$\text{Nyquist interval} = \frac{1}{\text{NR}} = \frac{1}{500} = 2 \text{ msec}$$